

Enrollment No./Seat No.:

GUJARAT TECHNOLOGICAL UNIVERSITY
Bachelor of Engineering - SEMESTER - IV EXAMINATION - WINTER 2025

Subject Code: 3140913

Date: 18-11-2025

Subject Name: Electrical Machine- I

Time: 02:30 PM TO 05:00 PM

Total Marks: 70

Instructions

- 1. Attempt all questions.**
- 2. Make suitable assumptions wherever necessary.**
- 3. Figures to the right indicate full marks.**
- 4. Simple and non-programmable scientific calculators are allowed.**

	Marks
Q.1 (a) State advantages and disadvantages of Swinburne's test.	03
(b) Explain principle of energy conservation.	04
(c) Derive an emf equation for transformer with usual notation.	07
Q.2 (a) Give classification of DC Motors with neat connection diagram.	03
(b) Compare electric and magnetic circuits.	04
(c) Analyze the O.C. and S.C. test on single phase transformer.	07
OR	
(c) Analyze the step by step voltage build up process of self-excited D.C. Generator.	07
Q.3 (a) Discuss the conditions to be satisfied before connecting two single-phase transformers in parallel.	03
(b) Differentiate between core type and shell type transformer.	04
(c) Derive the expression of armature torque developed in a dc motor using fundamental equation and power equation.	07
OR	
(a) State and explain the working principle of DC motor.	03
(b) Explain Scott-connection of transformer in detail.	04
(c) Explain the methods of speed control on d.c. shunt motor.	07
Q.4 (a) List the applications of d.c. motors.	03
(b) Explain singly excited magnetic field system in detail.	04
(c) A 4-pole long-shunt lap wound generator supplies 25 KW at a terminal voltage of 500 V. The armature, shunt field & series field resistances are 0.03Ω , 200Ω & 0.01Ω resp. Calculate the numbers of armature conductors if the speed of armature is 1200 RPM & flux/pole is 0.02 Wb.	07
OR	
(a) Explain Equalizer connection for DC machine.	03

- (b) Enlist different methods to improve commutation and explain any one method. 04
- (c) A 500 KVA Transformer has an iron loss and full load copper loss are 2.2 KW & 7.5 KW resp. Calculate the efficiency at (a) full load 0.8 p.f.(b) KVA load at which maximum efficiency occurs. 07

- Q.5**
- (a) Explain On Load Tap Changer for transformer. 03
 - (b) Explain 3-point starter for d.c. motor. 04
 - (c) Draw the vector and winding diagram for the following 3- phase transformer connections Dd0, Dy11, Yd1 . 07

OR

- (a) What is auto transformer ?, explain in brief. 03
- (b) Discuss series field test on D.C machine. 04
- (c) Draw the vector and winding diagram for the following 3- phase transformer connections Dd6, Dy1, Yd11. 07
